

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[])
{
    int source_fd, destination_fd;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    if (argc != 3)
    {
        printf("Usage: %s <source_file> <destination_file>\n", argv[0]);
        return 1;
    }

    // Open source file for reading
    source_fd = open(argv[1], O_RDONLY);

    if (source_fd < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    // Open/create destination file for writing
    destination_fd = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination_fd < 0)
    {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    // Read from source and write to destination
    while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0)
    {
        bytes_written = write(destination_fd, buffer, bytes_read);

        if (bytes_written != bytes_read)
        {
            perror("Error writing to destination file");
            close(source_fd);
            close(destination_fd);
            return 1;
        }
    }

    if (bytes_read < 0)
    {
        perror("Error reading source file");
    }

    // Close both files
    close(source_fd);
    close(destination_fd);

    printf("File copied successfully.\n");
}
```

```
adithya@LAPTOP-07A78KDI:~/OS$ nano practical2.c
adithya@LAPTOP-07A78KDI:~/OS$ gcc practical2.c -o practical2
adithya@LAPTOP-07A78KDI:~/OS$ ./practical2
Usage: ./practical2 <source_file> <destination_file>
adithya@LAPTOP-07A78KDI:~/OS$ echo "This is my Operating Systems practical." > source.txt
adithya@LAPTOP-07A78KDI:~/OS$ cat source.txt
This is my Operating Systems practical.
adithya@LAPTOP-07A78KDI:~/OS$
```

[illegible]

```

) = 24
sprintf(0, NULL, 8, NULL, 1048576, 0) = 0
close(0) = 0
close(0) = 0
close(0) = 0
signalstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size={0x19}, NULL}) = 0
munmap({curdir=/usr/local/lib64, 12288}) = 0
exit_group(0) = 7
+++ exited with 0 +++
adithya@LAPTOP-67A78MDI:~/OSS$ strace -e trace=openat,read,write,close cat sample.txt
open(AT_FDCWD, "/etc/lid.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
open(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
close(3) = 0
open(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
close(3) = 0
open(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
close(3) = 0
open(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
close(3) = 0
open(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libcprte-2.8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
close(3) = 0
open("/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbdev\tbd...", 1020) = 442
close(3) = 0
open(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
read(3, "none /usr/lib/modules/6.18.33.1-... ", 1024) = 1024
read(3, "e 0/nvmepts /dev/pts devpts rw,...", 1024) = 1024
read(3, "equid,nodew,noexec,relatime,nosr..., 1024) = 1024
read(3, "n/usr/1000 tmpfs rw,nosuid,node*..., 1024) = 1024
read(3, "...", 1024) = 0
close(3) = 0
open(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
read(3, "642730b0-1008-64273ba33000 --p 0*..., 1024) = 1024
read(3, "08 08:30 13645 .....", 1024) = 1024
read(3, "08 08:30 13648 .....", 1024) = 1024
read(3, "08 08:30 13648 /usr/lib/x86_64-lin..., 1024) = 1024
read(3, "08 13642 /usr/..., 1024) = 566
close(3) = 0
open(AT_FDCWD, "/usr/share/coreutils/locales/locale-en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
read(3, "%E common strings shared across A*..., 4080) = 4080
read(3, "...", 32) = 0
close(3) = 0
open(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 EONENT (No such file or directory)
open(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
Hello Operating Systems
close(0) = 0
close(0) = 0
close(0) = 0
+++ exited with 0 +++
adithya@LAPTOP-67A78MDI:~/OSS$ h
h: command not found
adithya@LAPTOP-67A78MDI:~/OSS |

```